

Lab 7 – Normalization

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1 Part one

1.1 Original table

PackageId	TagNumber	InstallDate	SoftwareCostUSD
AC01	32808	09-13-2005	754.95
DB32	32808	12-03-2005	380.00
	37691	06-15-2005	380.00
DB33	57772	05-27-2005	412.77
WP08	32808	01-12-2006	185.00
	37691	06-15-2005	227.50
	57222	05-27-2005	170.24
WP09	59836	10-30-2005	35.00
	77740	05-27-2005	35.00

1.2 (1) My professional consulting opinion

Tycho manufacturing's board needs to fire the CEO as a software company I expected better. Assuming this data is stored in a csv, excel, or some other raw text table structure to store the spreadsheet, the data is difficult to programmatically add, update, delete additional installs and packages. If an install needs to be deleted for whatever reason, the row with that install would have to have its internal structure of some columns edited to remove the correct install without affecting others. Additionally, nulls may also need to be left to store packages without installs. Accidental deletions of packages from deleting installs

could also be a problem. In my professional opinion, this data is not complex enough to warrant a database, but it would help organizing the spreadsheet more atomically.

1.3 (2) Spreadsheet in 1NF

PackageId	TagNumber	InstallDate	SoftwareCostUSD
AC01	32808	09-13-2005	754.95
DB32	32808	12-03-2005	380.00
DB32	37691	06-15-2005	380.00
DB33	57772	05-27-2005	412.77
WP08	32808	01-12-2006	185.00
WP08	37691	06-15-2005	227.50
WP08	57222	05-27-2005	170.24
WP09	59836	10-30-2005	35.00
WP09	77740	05-27-2005	35.00

1.4 (3) Primary key analysis

The best primary key that can be made with these columns is an composite key of the PackageID, TagNumber, and InstallDate. This has the side affect of disallowing multiple installs that have the same TagNumber and PackageID on the same day.

2 Part two adding Package Name and Computer Model

2.1 (4) New Table

PackageId	PackageName	TagNumber	ComputerModel	InstallDate	SoftwareCostUSD
AC01	Zork	32808	Lenova	09-13-2005	754.95
DB32	Roket Leauge	32808	Lenova	12-03-2005	380.00
DB32	Rocket Leauge	37691	Apple	06-15-2005	380.00
DB33	Chocolately	57772	IBM	05-27-2005	412.77
WP08	Neovim	32808	Lenova	01-12-2006	185.00
WP08	Neovim	37691	Apple	06-15-2005	227.50
WP08	Neovim	57222	IBM	05-27-2005	170.24
WP09	Balatro	59836	Dell	10-30-2005	35.00
WP09	Balatro	77740	MSI	05-27-2005	35.00

2.2 (5) Functional Dependencies

The functional dependencies are sets of columns where some column(s) are dependent on their value(s) from other column(s). In this table, PackageName is dependent on the PackageId and ComputerModel is dependent on TagNumber which makes these two functional dependencies.

2.3 (6) Non 3NF-ness

This table is not in third normal form because a part, importantly not the whole, of the primary key functionally determines another column. Both of the aforementioned functional dependencies create this problem as, for example, with just the PackageID you already know what the PackageName will be. Practically, this creates duplication within the table because you are storing identifying information of packages twice. You could just have another table which maps the PackageId to its PackageName. This way you would only need to write the PackageName once in the spreadsheet.

3 Part three

3.1 (7) Primary Keys

The primary keys for the tables to be split is the packageId, TagNumber, and TagNumber + PackageID respectfully.

3.2 (8) Functional dependencies

The functional dependencies for each table is only its primary key: Packages is PackageId, Installations is TagNumber and PackageId, finally Computers is TagNumber. These are the functional dependencies because the other columns only rely on those columns.

3.3 (9) 3NF

The tables are in third normal form because because there are no non-key or partial key functional dependencies. There are also no functional dependencies for any other columns onto another. The data also follows the previous forms of having no feilds with multiple items following a structure in just one column.

3.4 (10) Lucid Chart Diagram

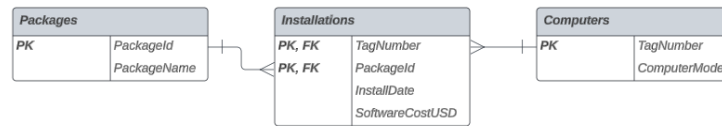


Figure 1: Crow's nest ER diagram